Sohee **Park**

Research Professor · AI Security Researcher

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Research Profile

Research professor and AI security researcher at the AI Safety Center, Soongsil University. Research focuses on AI model security and safety, privacy and trustworthy AI, generative AI security, and adversarial machine learning across vision, speech, language, and multimodal systems.

Education

Soongsil University — *Seoul, Republic of Korea*

Ph.D. in Software — *Mar. 2022 - Aug. 2025*

- Dissertation: *Multimodal-Based Defense Against Adversarial Attacks on Vision Models*.

Kongju National University — *Gongju, Republic of Korea*

M.Eng. in Convergence Science — *Mar. 2018 - Feb. 2020*

Kongju National University — *Gongju, Republic of Korea*

B.S. in Applied Mathematics — *Mar. 2014 - Feb. 2018*

Academic & Research Appointments

AI Safety Center, Soongsil University — *Seoul, Republic of Korea*

Research Professor — *Sep. 2025 - Present*

- Conducts research on AI model security and safety, privacy and trustworthy AI, generative AI security, and adversarial machine learning.
- Develops robust and proactive defenses across vision, speech, language, and multimodal systems.

Soongsil University — *Seoul, Republic of Korea*

Doctoral Researcher in Software — *Mar. 2022 - Aug. 2025*

- Developed multimodal defense methods for adversarial attacks on vision models.

Korea Education and Research Information Service (KERIS) — *Republic of Korea*

Researcher — *Jul. 2020 - Sep. 2021*

Teaching

2026	**Instructor, AI Security**, Soongsil University	*Spring*
2025	**Instructor, Big Data Analysis**, Soongsil University	*Fall*

Honors & Awards

2026	**Outstanding Paper Award**, ETRI President's Award, KIISC Chungcheong Branch Conference	*Korea*
2026	**Outstanding Paper Award**, NSR Director's Award, KIISC Summer Conference	*Korea*
2025	**Outstanding Paper Award**, ETRI President's Award, KIISC Summer Conference	*Korea*
2019	**Outstanding Paper Award**, ETRI President's Award, KIISC Winter Conference	*Seoul, Korea*
2017	**Outstanding Paper Award**, KIISC Summer Conference	*Seoul, Korea*

Research Interests

AI Model Security & Safety	Backdoors, red teaming, safety evaluation, and defenses for large language and other AI models
Privacy & Trustworthy AI	Personal-attribute inference, privacy leakage and suppression, and reliable AI evaluation
Generative AI Security	Detection and proactive defense against voice cloning, face manipulation, and other generative AI threats
Adversarial Machine Learning	Adversarial attacks, detection, and robust defenses across vision, speech, and multimodal systems

Technical Skills

Programming	Python
Machine Learning	PyTorch; model training and evaluation across language, vision, speech, and multimodal AI systems
AI Security Research	Adversarial attack and defense experiments, robustness and safety evaluation, and privacy-risk analysis
Languages	Korean (native); English (working proficiency with international conference presentation experience)

Academic Service

2027 **Conference Reviewer**, International Conference on Artificial Intelligence and Statistics (AISTATS)

2027 **Conference Reviewer**, International Conference on Learning Representations (ICLR)

2026 **Evaluations and Datasets Track Reviewer**, Conference on Neural Information Processing Systems (NeurIPS)

2026 **Reviewer, October Cycle**, ACL Rolling Review (ARR)

Selected Funded R&D Projects

2026 **Preventing Sensitive Information Inference and Risk Assessment in Foundation Model Operations**, Participating Researcher. KISA/PIPC grant (No. RS-2026-25526342; ongoing). Conducts research on sensitive-information inference prevention and privacy-risk evaluation for foundation models

2024–2025 **Countermeasure Technologies for Generative AI Security Threats**, Participating Researcher. IITP/MSIT grant (No. RS-2024-00398353). Conducted research on security threats and countermeasures for generative AI systems

2022–2025 **Robust AI and Distributed Attack Detection for Edge AI Security**, Operational Project Lead. IITP/MSIT grant (No. 2021-0-00511). Led project planning, administration, milestone and deliverable management, and research execution on robust AI and adversarial attack detection and defense

2022–2023 **User Identity Certification and Management for Self-Sovereign Identity Applications**, Participating Researcher. IITP/MSIT grant (No. 2021-0-00565). Conducted research on identity authentication and management technologies for self-sovereign identity

2021 **Defense R&D Project**, Participating Researcher. KRIT/DAPA grant (No. KRIT-CT-21-037). Conducted research as a participating researcher in a government-funded defense R&D project

2020 **Advanced Analysis and Defense of Adversarial-Example Vulnerabilities in AI**, Participating Researcher. NRF/MSIT grant (No. 2020R1A2C1014813). Conducted research on adversarial-example vulnerability analysis and defense technologies

2017–2019 **Remote Identity Verification and Anomalous-Transaction Filtering for Fintech Fraud Prevention**, Participating Researcher. Creative Seed Stage 2. Conducted research on machine-learning-based identity verification, fraud detection, and financial-risk prediction

Patent

2026 **Electronic Apparatus, Method, and Recording Medium for Generating an Adversarial Patch**, D. Choi, H. Na, W. Lee, S. Roh, and **S. Park**. U.S. Patent Application 19/218,242

Publications

Manuscripts Under Review

2027 **DART: Defense-Aware Red Teaming With a Critical Cue Bank for Open-World Adversarial Evaluation**, J. Shin, D. Min, J. Jang, **S. Park**, and D. Choi. Manuscript under review at the International Conference on Learning Representations (ICLR)

2026 **ImpliCue: A Benchmark for Implicit Personal Attribute Inference From Conversational Cues**, **S. Park**, C. Lee, W. Lee, M. Park, S. Ryu, and D. Choi. Manuscript under review

2026 **RoVo: Robust Voice Protection Against Voice Cloning Attacks via Embedding-Level Adversarial Perturbations**, S. Kim*, **S. Park***, D. Kim, J. Lee, and D. Choi. Manuscript under review. *Equal contribution

International Journal Articles

2026 **Multi-View Self-Supervised Learning for Label-Flipping Robustness in Tabular Data: A Comparative Study**, J. Jang, **S. Park**, S. S. Nam, and D. Choi. Applied Intelligence, 56(15), 466

2026 **TRACER: Temporal Retrieval-Augmented Contextual Evaluator for Robustness in ML Training**, J. Shin, J. Jang, **S. Park**, J. Lee, and D. Choi. Expert Systems with Applications, Article 133668

2025 **Detecting Adversarial Examples Using Cross-Modal Semantic Embeddings From Images and Text**, **S. Park**, G. Ryu, and D. Choi. IEEE Access, 13, 39330-39342

Domestic Journal Articles

2025 **Research Trends in Suppressing Harmful Content Generation by Stable Diffusion**, J. Shin, **S. Park**, and D. Choi. Review of KIISC, 35(3), 79-96

2024 **Detection of Synthetic Speech Using Language-Model-Based Speech Feature Extraction**, S. Kim, **S. Park**, and D. Choi. Journal of the Korea Institute of Information Security & Cryptology, 34(3), 439-449

2024 **AI Security Vulnerabilities in Fully Unmanned Stores: Adversarial Patch Attacks and Data-Augmentation Defense**, W. Lee, H. Na, **S. Park**, and D. Choi. Journal of the Korea Institute of Information Security & Cryptology, 34(2), 245-261

2023 **Membership Inference Attack Against Text-to-Image Models Using Adversarial Prompts and Textual Inversion**, Y. Oh, **S. Park**, and D. Choi. Journal of the Korea Institute of Information Security & Cryptology, 33(6)

2022 **Adversarial Example Detection Based on Symbolic Representation of Images**, **S. Park**, S. Kim, H. Yoon, and D. Choi. Journal of the Korea Institute of Information Security & Cryptology, 32(5), 975-986

2020 **Improving Imbalanced Numerical Data Using Tomek Links Combined With Balancing GAN**, H. Na, **S. Park**, and D. Choi. Journal of KIISE, 47(10), 974-984

2020 **A User Authentication Model Using Web-Standard Device Fingerprints**, **S. Park**, J. Jang, and D. Choi. Journal of the Korea Institute of Information Security & Cryptology, 30(4), 631-646

2019 **Loan-Status Analysis and Default-Prediction Performance Using a Personal-Credit Sample Database**, **S. Park** and D. Choi. Journal of KIISE, 46(7), 627-635

2019 **Machine-Learning-Based Prediction of P2P Lending Deadlines**, **S. Park** and D. Choi. Journal of KIISE, 46(2), 174-183

2017 **Artificial Intelligence Security Issues**, **S. Park** and D. Choi. Review of KIISC, 27(3), 27-32

International Conference & Workshop Papers

2026 **Reducing Logprob Without Causal Verification: Diagnosing SAE-Based PII Suppression in LLMs**, C. Lee, W. Lee, **S. Park**, and D. Choi. Findings of the Association for Computational Linguistics: The 5th Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics and the 15th International Joint Conference on Natural Language Processing (AACL-IJCNLP)

2026 **NaVo: Natural Voice Protection Against Voice Cloning Attacks via Generative Universal Adversarial Audio**, S. Park, S. Kim, **S. Park**, D. Kim, T. A. Nguyen, T.-P. Doan, S. Jung, and D. Choi. Interspeech

2026 **SemantiGuard: Explainable Adversarial Detection via Semantic Similarity of Knowledge**, G. Kim, **S. Park**, and D. Choi. 22nd International Conference on Advanced Visual and Signal-Based Systems (AVSS)

2026 **Event-Triggered Decoupled MARL for Real-Time Swarm UAV Control Under Embedded Constraints**, S. Roh, **S. Park**, and D. Choi. 26th IEEE/ACM International Conference on Embedded Software (EMSOFT)

2026 **STAR: Detecting Inference-Time Backdoors in LLM Reasoning via Prior-Posterior Amplification**, S. G. Park*, **S. Park***, J. Lee, H. Na, and D. Choi. Findings of the Association for Computational Linguistics: Empirical Methods in Natural Language Processing (EMNLP). *Equal contribution

2026 **RoCo: Robust Code for Fast and Effective Proactive Defense Against Voice Cloning Attack**, S. Kim, D. Kim, **S. Park**, and D. Choi. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)

2026 **TRACE: Confounder-Free Adversarial Fine-Tuning for Robust Object Detection**, W. Lee, J. Lee, H. Na, **S. Park**, and D. Choi. IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)

2026 **ASTRA: Adversarial Stealthy Trigger Reasoning Attacks for Black-Box LLMs**, S. G. Park, **S. Park**, and D. Choi. Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD), pp. 578-589

2025 **Head Feature-Based Adversarial Attack for Proactive Defense Against Face Swapping**, D. Kim, **S. Park**, S. Roh, and D. Choi. IEEE International Conference on Advanced Video and Signal-Based Surveillance (AVSS)

2025 **Infrared Thermal-Guided Adversarial Patch Defense for Robust Visible Person Detectors**, **S. Park**, H. Jeong, and D. Choi. IEEE International Conference on Advanced Video and Signal-Based Surveillance (AVSS)

2018 **Active Authentication Experiments Using Actual Application-Usage Logs**, G. Ryu, **S. Park**, D. Choi, Y. Kim, S. H. Kim, S. Kim, D. Kim, and D. Kwon. Workshop on Radical and Experiential Security, pp. 9-16

Domestic Conference Papers

2017 **Loan-Review Performance Experiments Using a Multi-Layer Perceptron**, **S. Park** and D. Choi. Proceedings of the Korea Computer Congress, pp. 1899-1900